

A Cofinal Finsler--Bezoutian Completion Criterion for the Riemann Hypothesis

Riemann Collaborative Research Project | 31 July 2026

STATUS. This note proves a conditional finite-to-global theorem and two equivalent finite criteria. It does not prove that the required cofinal hypotheses hold, and therefore does not claim a proof of RH.

Abstract

Let f_j be exact finite Fourier transforms converging locally uniformly to $\Xi(z) = \xi(1/2 + iz)$ on the open strip $|\operatorname{Im} z| < 1/2$. If, at every sufficiently large level, an exact target-pinned Connes--van Suijlekom special matrix is positive semidefinite with one-dimensional even kernel generated by the target, then every f_j is real-rooted. Hurwitz then excludes nonreal zeros of Ξ in the strip, hence proves RH. The finite positivity condition is equivalent to positivity of a pinned Weil form on the isotropic cone of a universal slope form; in the simple-root regime it is equivalent to strict separation of finitely many Bezoutian root thresholds.

1. Main theorem

For every j let p_j be a nonzero real even finite target and put

$$f_j(z) = \hat{(p_j)}(z).$$

Assume $f_j \rightarrow \Xi$ locally uniformly on $S = \{z : |\operatorname{Im} z| < 1/2\}$.

Let η_j be the boundary vector, $\delta_j = \eta_j^T p_j \neq 0$, and

$$p_j = p_j / \delta_j, \quad \eta_j^T p_j = 1.$$

Let Q_j be the exact real symmetric special matrix in the same nodes and normalization. Define

$$\begin{aligned} T_j(c) &= A_j + c B_j, \\ A_j &= Q_j - \operatorname{diag}((Q_j p_j)_i / (p_j)_i), \\ B_j &= \operatorname{diag}((\eta_j)_i / (p_j)_i) - \eta_j \eta_j^T. \end{aligned}$$

Then $A_j p_j = B_j p_j = T_j(c) p_j = 0$ for every real c .

Theorem

If, for every sufficiently large j , there is c_j in $\mathbb{Q}$ such that

$$T_j(c_j) \succeq 0, \quad \ker T_j(c_j) = \mathbb{R} p_j,$$

then the Riemann Hypothesis is true.

2. Finite real-rootedness and the Hurwitz passage

In the exact finite coordinates, the off-diagonal entries of Q_j have divided-difference form $(\beta_r - \beta_s)/(\lambda_r - \lambda_s)$. The scalar update subtracts c from every off-diagonal entry and therefore replaces β_r by $\beta_r - c$ λ_r . The forced diagonal is allowed by the finite special-matrix theorem, and parity is preserved.

Thus $T_j(c_j)$ is a positive semidefinite special matrix with a one-dimensional even kernel. The finite Connes--van Suijlekom theorem implies that every zero of $\hat{p}_j$ is real.

Because $f_j = \delta_j \hat{p}_j$ with $\delta_j \neq 0$, every f_j has the same real zero set.

Hurwitz argument

The upper and lower half-strips $S^+ = \{0 < \text{Im } z < 1/2\}$ and $S^- = \{-1/2 < \text{Im } z < 0\}$ are connected. Every sufficiently large f_j is zero-free on each half-strip. Local uniform convergence and Hurwitz imply that ξ_i is either zero-free or identically zero there. ξ_i is not identically zero (for example $\xi_i(2) = \pi/6$), hence ξ_i is zero-free on S^+ union S^- .

If $\rho = \beta + i\gamma$ is a nontrivial zeta zero, then

$$\begin{aligned} z_\rho &= (\rho - 1/2)/i = \gamma - i(\beta - 1/2), \\ \xi(z_\rho) &= \xi(\rho) = 0. \end{aligned}$$

Since $0 < \beta < 1$, z_ρ lies in S . The half-strip conclusion forces z_ρ to be real, so $\beta = 1/2$. Therefore every nontrivial zero lies on the critical line.

3. Exact Finsler equivalence

Fix one level, set $H = p^\perp$, and write

$$a(x) = x^T A_p x, \quad b(x) = x^T B_p x.$$

Since $A_p p = B_p p = 0$, the matrix condition is exactly

$$a(x) + c b(x) > 0 \quad \text{for every nonzero } x \text{ in } H.$$

The restricted slope $B_p|_H$ is nonsingular. Strict Finsler/Dines separation gives

$$\begin{aligned} &\text{exists } c \text{ with } A_p + c B_p > 0 \text{ on } H \\ &\text{iff } a(x) > 0 \text{ whenever } x \text{ in } H \setminus \{0\} \text{ and } b(x) = 0. \end{aligned}$$

This includes the definite-slope cases, where the nonzero isotropic cone is empty and a sufficiently large scalar of the appropriate sign works.

4. Feasible scalar interval and rational certificates

Define

$$c_- = \sup_{\{x \in H, b(x) > 0\}} -a(x)/b(x),$$

$$c_+ = \inf_{\{x \in H, b(x) < 0\}} -a(x)/b(x).$$

The complete feasible set is the open interval

$$c_- < c < c_+.$$

The isotropic-cone condition is equivalent to $c_- < c_+$. Because the interval is open, every analytic success admits a rational certificate c . Conversely, one rational c plus exact positive LDL pivots on $p^\perp$ is a complete finite proof object.

5. Simple-root Bezoutian form

Let the distinct real nodes be λ_i and set

$$\Omega(s) = \prod_i (\lambda_i - s), \quad \phi_i(s) = \Omega(s) / (\lambda_i - s),$$

$$P(s) = \sum_i p_i \phi_i(s).$$

For the completed source define

$$R_c(s) = \sum_i p_i (\beta_i - c \lambda_i) \phi_i(s),$$

$$B_c(s, t) = [P(s)R_c(t) - P(t)R_c(s)] / (s - t).$$

The node-evaluation matrix of this Bezoutian is congruent, with a fixed minus sign, to $T_p(c)$. If P has simple real roots r_k , root evaluation diagonalizes the nonzero Bezoutian into $P'(r_k)R_c(r_k)$. Hence

$$T_p(c) \succeq 0, \quad \ker T_p(c) = R_p$$

$$\text{iff } P'(r_k)R_c(r_k) < 0 \text{ for all } k.$$

At a target root, $R_c(r_k) = R_0(r_k) - c \Omega(r_k)$. Put

$$c_k = R_0(r_k) / \Omega(r_k), \quad \sigma_k = P'(r_k) \Omega(r_k).$$

The exact finite condition is therefore

$$\max_{\{\sigma_k > 0\}} c_k < \min_{\{\sigma_k < 0\}} c_k.$$

Empty sides are interpreted as infinite. Repeated roots require a confluent Bezoutian/Hermite criterion and are not covered by this scalar root formula.

6. Logical chain and proof boundary

target convergence + cofinal finite completion
=> finite real-rooted transforms
=> ξ zero-free off the real axis in $|\operatorname{Im} z| < 1/2$
=> RH.

Equivalent cofinal finite hypotheses

At every sufficiently large level it is enough to prove either

$$x^{\wedge T} A_{\{p_j\}} x > 0 \text{ whenever } x \perp p_j, x \neq 0, x^{\wedge T} B_{\{p_j\}} x = 0,$$

or, in the simple-real-root regime,

$$\max_{\{\sigma_{j,k} > 0\}} c_{j,k} < \min_{\{\sigma_{j,k} < 0\}} c_{j,k}.$$

What remains unproved

This note does not prove that exact repaired Hermite-radical targets satisfy the required local-uniform convergence in the final production normalization, nor that the Finsler/Bezoutian separation holds cofinally. Those are the substantive remaining global estimates. The theorem here closes the finite-to-global implication and proves the equivalence of the proposed finite positivity criteria.

References

- [1] A. Connes and W. D. van Suijlekom, Quadratic Forms, Real Zeros and Echoes of the Spectral Action, arXiv:2511.23257.
- [2] P. Finsler, Ueber das Vorkommen definiter und semidefiniter Formen in Scharen quadratischer Formen, 1936.
- [3] L. L. Dines, On the mapping of quadratic forms, Bulletin of the AMS 47 (1941), 494-498.
- [4] Classical Hurwitz theorem for locally uniform limits of zero-free holomorphic functions.

Repository status: conditional theorem proved under stated hypotheses; RH itself is not claimed.